

Skills Summary

Dilations and Similar Figures

Use this reference while completing your worksheet or when studying.

A dilation multiplies every length by the scale factor k and leaves every angle alone. That is why the image is *similar* to the original — same shape, different size. A dilation is the one transformation on this sheet that is not rigid.

Skill 1 — Coordinates of a dilated point

About the origin: $(x, y) \rightarrow (kx, ky)$.

About a centre (h, j) : $(x, y) \rightarrow (h + k(x - h), j + k(y - j))$ — measure the offset from the centre, scale the offset, add it back.

$k > 1$ enlarges, $0 < k < 1$ shrinks, and the centre is the one point that never moves.

Example: Dilate $(-4, 5)$ about the origin by a factor of 3.

Step 1. The centre is the origin, so each coordinate can be scaled on its own — no offsets are needed.

Step 2. $x' = 3(-4)$ and $y' = 3(5)$
 $\Rightarrow (-12, 15)$

Try it: Dilate $(7, 4)$ about $(1, 1)$ by a factor of 3 and give the x -coordinate of the image.

check: 61

Skill 2 — Finding the scale factor from two lengths

Rule: $k = \frac{\text{image length}}{\text{original length}}$ — a division of matching sides, never a subtraction.

Convert both lengths to the same unit first. A model in centimetres against a real object in metres gives a scale factor a hundred times too small if the conversion is skipped.

Example: A 6 cm side becomes 21 cm after an enlargement. Find the scale factor.

Step 1. A scale factor answers “how many times as long”, and that is a ratio, so divide the new length by the old one rather than finding the gap between them.

Step 2. $k = 21/6$
 $\Rightarrow k = 3.5$

Try it: An 18 cm side becomes 45 cm. Find the scale factor.

check: 2.5

Skill 3 — Similar figures in real situations

Set up a proportion of matching parts: $\frac{\text{height}_1}{\text{shadow}_1} = \frac{\text{height}_2}{\text{shadow}_2}$, and cross-multiply.

Keep both fractions built the same way round — height over shadow on both sides, or shadow over height on both sides. Mixing them is the commonest error in this topic.

Scale drawings work the same way: $\text{real length} = \text{drawn length} \times \text{the scale}$.

Example: A 2 m post casts a 3 m shadow; a tree casts a 9 m shadow. Find the tree's height.

Step 1. The post with its shadow and the tree with its shadow form similar right triangles, so matching sides are in the same ratio — set up height over shadow on both sides.

Step 2. $\frac{h}{9} = \frac{2}{3}$, so $h = \frac{2 \times 9}{3}$
 $\Rightarrow h = 6 \text{ m}$

Try it: A 1.8 m post casts a 4 m shadow; a tree casts a 20 m shadow. Find the tree's height in metres.

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(!) Watch out: a scale factor k multiplies lengths by k , but it multiplies *areas* by k^2 . Doubling every side of a rectangle quadruples its area — so never apply a linear scale factor straight to an area.